Telegram as a Transformative Criminal Marketplace: Analyzing the Implications of Identity Theft Dynamics on Surface-Accessible Platforms

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Abstract

Purpose: Messaging platforms like Telegram have become key marketplaces for stolen personally identifiable information (PII), transforming how such data is organized, distributed, and monetized. Guided by Routine Activity Theory, we explore: (1) how Telegram's architecture enables the convergence of offenders, targets, and weak guardianship; (2) which types of data are most frequently shared; and (3) how victimization patterns vary by geography and institution. This research addresses a critical gap in understanding how surface-accessible messaging platforms have transformed identity theft operations, moving beyond darknet-focused cybercrime studies.

Methods: We developed a multistage framework to explore Telegram's role in identity theft. Channels were identified via keyword searches and network exploration, followed by automated data collection. Extracted posts were processed to convert unstructured content into usable intelligence. This enabled tracking of data flows across 25 Telegram channels throughout 2023, offering visibility into underground markets typically hidden from traditional investigations.

Results: Credit card data dominated, comprising 82% of all shared PII, highlighting strong financial incentives. Geographic analysis showed unexpected concentrations in Texas, California, and Florida. Temporal spikes in March and June aligned with consumer spending peaks, suggesting strategic timing by offenders.

Conclusion: These findings update Routine Activity Theory by demonstrating how Telegram creates a unique "criminal convergence space" where motivated offenders, suitable targets, and absent guardians interact. Unlike darknet markets, Telegram's low barrier to entry and anonymity features facilitate identity theft while impeding enforcement. This study reveals new cybercrime dynamics and supports future development of real-time monitoring, victim alert systems, and predictive risk models.

Keywords: Telegram, Instant Messaging Platforms, PII, Identity Theft, Financial Fraud, Cybercrime.

1 Introduction

As digital communication becomes ubiquitous, cybercriminals have adapted their tactics by exploiting emerging technologies for illicit financial gain. Identity theft, one of the most pervasive forms of cybercrime, has surged in recent years due to the widespread accessibility of personal information online.

In 2021 alone, identity theft affected over 23 million individuals in the United States, with card theft accounting for 7.2 million cases and financial losses exceeding $16 billion (Victims of Identity Theft, 2021) [1]. Broader forms of direct-contact identity fraud affected more than 27 million people, contributing to a cumulative loss of $28 billion (Javelin Strategy and Research, 2022) [2]. Given the alarming frequency of such incidents, where cybersecurity experts estimate a new victim emerges every 22 seconds (Identity Theft Facts and Statistics, 2024) [3], it is critical to understand how this data is obtained, circulated, and monetized.

Understanding the mechanisms behind identity theft requires examining the underlying supply chains that facilitate the movement of stolen personal information from initial compromise to criminal exploitation. These criminal ecosystems have evolved sophisticated organizational structures that mirror legitimate business operations, with clearly defined roles for suppliers, processors, distributors, and end-user retailers of compromised credentials. Traditional cybercriminal operations have relied on structured darknet marketplaces that require specialized technical knowledge and follow linear supply chain models similar to conventional commerce (Soska and Christin, 2015; Hutchings and Holt, 2015) [4, 5].

However, a significant shift has occurred in how cybercriminals obtain and distribute stolen data, with many operations moving from specialized darknet marketplaces to more accessible messaging platforms like Telegram. This migration represents

not just a change in criminal tactics or supply chain organization, but a fundamental shift in how the elements of crime converge, a transformation we examine through the lens of Routine Activity Theory (Cohen and Felson, 1979) [6].

Routine Activity Theory proposes that crime occurs when three elements converge: (1) a motivated offender, (2) a suitable target, and (3) the absence of capable guardianship. Telegram's architecture creates optimal conditions for this convergence by providing a user-friendly accessible space where motivated cybercriminals can easily access suitable targets while evading traditional forms of guardianship. Unlike darknet markets that require technical expertise to access (creating a form of guardianship through complexity), Telegram's combination of accessibility and anonymity creates what we term a "Transformative Criminal Convergence Space" that facilitates and regularizes participation in cybercrime while maintaining protection from law enforcement intervention.

1.1 The Need for Systematic Study of Digital Criminal Environments

Traditional methods of monitoring cybercriminal activities developed for darknet forums and static repositories, are inadequate for dynamic messaging platforms for several reasons: the volume and velocity of information exchanged in these messaging platforms exceed manual analysis capabilities; the ephemeral nature of communications (with rapid deletion or channel migrations) creates significant challenges for historical analysis; and the use of specialized jargon and coded communications requires contextual understanding to properly interpret.

Developing frameworks for systematic study of these criminal hubs requires methodologies that can capture both the content and patterns of communication at scale. This includes identifying active channels, documenting the types and formats of information being exchanged, analyzing the geographic and institutional distribution of compromised credentials, and tracking temporal patterns that may correlate with external events or criminal operational cycles.

By developing a broad understanding of how these criminal ecosystems function, researchers and security professionals can identify vulnerabilities within criminal operations, anticipate emerging threats, and develop more effective countermeasures. This understanding also provides the foundation for theoretical advancement in criminology, particularly in how digital platforms transform criminal opportunities and behaviors.

1.2 Current Approaches to Identity Theft Impact Mitigation and Their Limitations

Current strategies to mitigate identity theft rely primarily on reactive measures that activate after credentials have already been compromised (stolen from legitimate sources and exposed to unauthorized access). Financial institutions and credit card

companies offer monitoring services that alert customers to unusual activity on their accounts or changes in their credit reports (Credit Monitoring Service 2024) [7]. All 50 states in the United States have enacted laws mandating financial companies to notify customers if their personal information is compromised in a data breach and to recommend protective measures such as fraud alerts or credit freezes (Security Breach Notification Laws 2022) [8].

At the federal level, the Fair and Accurate Credit Transactions Act (FACTA) established a system of fraud alerts and identity theft prevention practices (Federal Trade Commission 2003) [9]. Various government agencies, including the Federal Trade Commission (FTC), U.S. Department of Justice (DOJ), and Internal Revenue Service (IRS), collaborate with state agencies to investigate, prosecute, and prevent identity theft.

However, these measures face significant limitations in addressing identity theft facilitated through instant messaging platforms. First, they operate on a post-incident basis, creating a window of vulnerability between initial data compromise and detection. Second, they lack visibility into the criminal ecosystems where stolen credentials are traded, limiting their ability to anticipate emerging threats or patterns. Third, they operate primarily at the individual level, making it difficult to identify broader patterns or systemic vulnerabilities that could inform more comprehensive protective measures.

Commercial identity theft protection services, such as Discover Identity Theft Protection and similar offerings from LifeLock, Aura, and other providers, represent current attempts to address some of these limitations through more comprehensive monitoring approaches (Discover Identity Theft Protection 2024) [10]. These services typically provide three-bureau credit monitoring, dark web scanning for compromised personal information, and real-time alerts when suspicious activity is detected (Security.org 2025; U.S. News 2025) [11, 12]. Services like Discover offer monitoring of Social Security numbers, banking information, and criminal court records, along with up to $1 million in identity theft insurance coverage and dedicated fraud resolution specialists (Aura 2024) [13]. However, even these advanced commercial systems remain fundamentally reactive, activating only after personal information has already been compromised and circulated within criminal networks. They cannot provide early warning of credential exposure on platforms like Telegram, where stolen information is actively traded before traditional monitoring systems can detect its misuse.

1.3 Contributions of This Research

Given these significant limitations in current identity theft mitigation strategies and the gaps in understanding how criminals operate on platforms like Telegram, this research addresses three critical questions derived from our investigation.

- First, we explore how Telegram's architecture enables the convergence of offenders, targets, and weak guardianship in new ways, establishing a framework for

understanding how surface-accessible messaging platforms are reshaping criminal ecosystems.
- Second, we explore which types of data are most frequently shared on Telegram channels focused on identity theft. Understanding which forms of Personally Identifiable Information (PII) are prioritized provides insight into criminal preferences and financial motivations within these ecosystems.
- Third, we analyze how victimization patterns vary by geography and institution, examining targeting strategies across different banks, card types, and geographic regions. These patterns reveal concentrations of criminal activity that can inform protective measures.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on online illicit marketplaces, the role of messaging platforms in facilitating cybercrime, PII collection, and digital crime spaces. Section 3 outlines our research methodology for studying Telegram channels engaged in identity theft activities and defines our framework. Section 4 describes the implementation of our data collection and extraction workflow. Section 5 presents the results of our analysis, addressing how Telegram's architecture enables criminal convergence, which types of data are most frequently shared, and how victimization patterns vary by geography and institution. Section 6 discusses the implications of our findings, including theoretical contributions, practical applications, and methodological considerations. Section 7 addresses the limitations of this research. Finally, Section 8 concludes the research and discusses potential avenues for future investigation.

2 Literature Review

Research on illicit online marketplaces, particularly those trafficking financial data and other contraband, has grown significantly over the past decade. Early studies often focused on dark web forums, revealing the intricate networks and supply chains that support the sale of stolen information. More recent work has begun to examine surface-accessible messaging platforms like Telegram, where cybercriminals can find a way around law enforcement investigations on traditional dark web marketplaces.

2.1 Evolution of Online Illicit Marketplaces and Criminal Supply Chains

Understanding criminal supply chains requires examining their relationship to legitimate business operations that serve as organizational models for illicit activities. Traditional industrial supply chains follow well-established linear patterns where raw materials flow through sequential stages of processing, distribution, and retail delivery to end consumers (Porter 1985; Christopher 1992) [14, 15]. This conventional model features clearly defined roles including suppliers who provide raw materials,

manufacturers who transform materials into finished products, distributors who manage logistics and warehousing, and retailers who facilitate final consumer transactions (Cooper and Ellram 1993; Lambert and Cooper 2000) [16, 17] (Fig. 1).

Fig. 1 Traditional Industrial Supply Chain

Drawing from these established business frameworks, criminal organizations have historically adapted similar structural approaches to organize their illicit operations. Traditional darknet marketplaces operated through structured models resembling legitimate business operations, with clearly defined sequential stages mirroring conventional industrial processes shown in Fig. 1 (Fig. 2). This criminal adaptation of the industrial supply chain begins with hackers (suppliers) who exploit vulnerable systems to steal sensitive information. The stolen data is then processed and packaged by data processors (manufacturers) who format it for distribution. Marketplace operators (distributors) subsequently manage the logistics of advertising and selling bulk datasets through specialized platforms, while individual sellers (retailers) facilitate final transactions with end-user criminals (consumers) through darknet websites accessible only via specialized browsers like Tor.

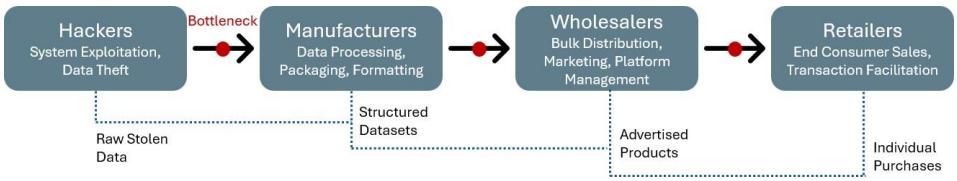

Fig. 2 Traditional Underground Supply Chain

This linear model created bottlenecks and dependencies between each stage. For example, if hackers fail to steal sufficient data, manufacturers have nothing to process; if key marketplaces are shut down by law enforcement, the entire distribution network collapses; if technical infrastructure fails or cryptocurrency exchanges are compromised, all transactions halt immediately. This makes the entire operation vulnerable to disruption at key points.

However, existing literature provides limited analysis of how newer platforms like Telegram may be transforming these traditional supply chain models. Recent studies by Garkava et al. (2024) [18] and Wang et al. (2024) [19] document criminal activities on Telegram but do not provide comprehensive analysis of supply chain evolution. This gap in understanding the transformation from linear to potentially more fluid criminal supply chain models represent a critical area for investigation.

Early foundational studies by Holt and Lampke (2010) [20], Motoyama et al. (2011) [21], and Hutchings and Holt (2015) [5] examined the structure and operations of cybercriminal forums, documenting how stolen data markets function as economic systems with specialized roles, pricing structures, and trust mechanisms. These early studies established critical frameworks for analyzing the linear supply chain model illustrated in Fig. 2, focusing primarily on forum-based platforms that operated through clearly defined sequential stages from data theft to end-user transactions.

Building upon this foundation, more recent research has examined the economic dynamics of these markets. Ablon et al. (2014) [22] conducted a comprehensive analysis of cybercriminal markets, detailing various goods and services available, including financial information, malware tools, and hacking services. Their work highlights how these markets operate as specialized economic ecosystems with defined roles and market-driven pricing structures characteristic of the linear model shown in Fig. 2. Similarly, Holt et al. (2016) [23] investigated the international landscape of illicit financial data markets, emphasizing forums and Internet Relay Chat (IRC) channels as operational hubs. These studies collectively establish the economic foundations and operational practices of the sequential supply chain model but preceded the significant shift toward messaging platforms that may challenge this linear structure.

The evolution of digital black markets has prompted researchers to explore increasingly dynamic environments. Soska and Christin (2015) [4], D´ecary-H´etu and Giommoni (2017) [24], and Jardine (2019) [25] documented the rise and fall patterns of darknet marketplaces, highlighting their vulnerability to law enforcement interventions and the resulting market instability. These studies provide valuable insight into the structural weaknesses inherent in the linear supply chain model, particularly the single points of failure that make entire networks vulnerable to disruption. Their findings help explain why criminals might seek alternative platforms with enhanced security features and operational flexibility, setting the stage for understanding the potential migration away from traditional sequential supply chains toward more resilient operational models.

2.2 Emergence of Instant Messaging Platforms in Cybercrime

The shift from specialized darknet markets to surface-accessible messaging platforms as an evolution in cybercriminal operations has only recently attracted scholarly attention. Europol's Internet Organised Crime Threat Assessment (2019, 2020, 2021) [26] has consistently identified social messaging applications as emerging vectors for criminal activity, noting their growing popularity among cybercriminals seeking

to evade traditional law enforcement monitoring. These reports provide important institutional recognition of the trend but offer limited empirical analysis of how these platforms transform criminal operations.

Hardy and Norgaard (2016) [27] and Broadhurst et al. (2021) [28] have each documented various aspects of criminal operations shifting to instant messaging platforms, including drug trafficking, fraud schemes, and contraband sales. These studies highlight how such platforms offer criminals a combination of security features and operational flexibility that traditional darknet markets cannot match, though they focus primarily on narcotics trafficking rather than identity theft operations.

Garkava et al. (2024) [18] have conducted a comprehensive crime script analysis of stolen data markets on Telegram, documenting how the platform has become a hub for trading PII and financial data across 16 observed channels. Their study offers valuable insights into the operational dynamics of these markets and proposes situational crime prevention measures to disrupt them. Wang et al. (2024) [19] provide the first large-scale analysis of cybercriminal activity channels on Telegram, examining 339 channels followed by over 23.8 million users that distribute compromised credentials, hacking tools, and malware. Additionally, Bekkers et al. (2024) [29] have documented how cybercriminal networks increasingly use Telegram and Snapchat to recruit co-offenders for financial fraud operations. These pioneering studies establish that Telegram has become a significant platform for cybercriminal operations while offering detailed insights into the specific dynamics of identity theft activities within these digital spaces. However, existing research lacks a comprehensive framework for understanding how Telegram's unique architectural features enable criminal convergence, systematic analysis of which types of personal information are most frequently traded, and empirical examination of geographic and institutional targeting patterns that could inform protective measures.

2.3 PII Collection and Analysis Methodologies

Parallel to analyses of cybercriminal markets, other studies have advanced techniques for collecting and analyzing PII exposed through various channels.

Automated scanning and collection methods have been documented by several researchers. Thomas et al. (2017) [30] and Liu et al. (2020) [31] developed systems for monitoring data breach disclosures and scanning dark web forums to identify exposed PII. These studies pioneer techniques for large-scale automated collection but typically focus on static repositories rather than the dynamic, temporary nature of messaging platform communications. Similarly, Po-Yi Du et al. (2018) [32] and Benjamin et al. (2015) [33] proposed approaches for gathering cyber threat intelligence from hacker communities using custom web crawlers and parsers, establishing important methodological precedents for analyzing criminal communications.

More recent methodological innovations have specifically targeted encrypted or semi-encrypted platforms. Zaeifi et al. (2024) [34] explored the potential of identity fraud within financial ecosystems by examining PII-based authentication across

financial services and evaluating cost-benefit dynamics for criminals. Their multistage methodology demonstrates how researchers can systematically analyze PII vulnerability across different contexts. Similarly, Lin et al. (2020) [35] employed advanced machine learning techniques to link PII across surface and dark web environments, while Liu et al. (2021) [36] introduced transfer learning methodologies to extract multiple forms of PII from social media posts.

Despite these important methodological advances, there remains a gap in approaches designed for messaging platforms like Telegram. The unique characteristics of these platforms, including their hybrid public/private nature, transitory messages, and multimedia content, create methodological challenges that are not fully addressed by current approaches. Additionally, most existing methodologies focus on identification and collection rather than pattern analysis across time, geography, and institutional targets.

2.4 Theoretical Frameworks for Digital Crime Spaces

Several theoretical frameworks have been employed to understand cybercriminal behaviors and operational patterns. These frameworks provide important conceptual tools for our analysis while highlighting the need for theoretical advancement specific to surface-accessible messaging environments.

Routine Activity Theory (Cohen and Felson 1979) [1] has been adapted to digital contexts by several researchers, including Leukfeldt and Yar (2016) [37], Maimon et al. (2013) [38] and Reyns et al. (2011) [39].These adaptations examine how digital environments reshape the convergence of motivated offenders, suitable targets, and capable guardians, establishing important precedents for our concept of "criminal convergence space". However, these applications typically focus on general cybercrime patterns rather than specific platform dynamics.

Other theoretical frameworks have examined the structural characteristics of digital platforms that enable or constrain criminal activities. Norman's (1988) [40] concept of affordances, extended to digital environments by Hutchby (2001) [41] and subsequently applied to cybercrime by Shaw (2019) [42] and Europol (2021) [43], provides tools for analyzing how platform features shape criminal opportunities. Similarly, Calkins et al. (2015) [44] and Wall (2007) [45] have explored how digital platforms create distinct criminal ecosystems with unique operational patterns.

Despite these important theoretical contributions, there remains limited integration of these frameworks with empirical analysis of messaging platforms. Our concept builds upon these foundations but requires empirical validation through systematic analysis of criminal operations within these environments.

3 Research Methodology

This section outlines the structured, multi-stage methodology to systematically identify, collect, and analyze personal and financial data shared on Telegram channels associated with identity theft. Our approach was designed to address the unique features of surface-accessible messaging platforms, balancing data quality, scalability and ethical considerations while contributing to a replicable model for future cybercrime research.

3.1 Research Design Framework

Our methodology employs a five-stage data collection and analysis framework that builds upon established methodologies (refer to Section 2.3) for studying criminal ecosystems in surface-accessible messaging platforms while addressing the unique challenges presented by Telegram's architecture and the nature of identity theft operations. This framework directly addresses our three research questions (RQs) through integrated methodological components. The overall methodology is visualized in Fig. 3, which outlines the sequential pipeline: beginning with the identification of relevant Telegram channels, followed by automated data scraping, structured data extraction and parsing, validation and quality checks, and ending with typological, geographic, and institutional analyses of the collected data. RQ1 is addressed through Stages 1, 2, and 5 by examining how Telegram's architecture enables criminal convergence and operational dynamics. RQ2 is targeted through Stages 3 and 4 via data type categorization, frequency analysis, and validation procedures. RQ3 is investigated across Stages 3, 4, and 5 through geographic and institutional pattern analysis of victimization data.

Stage 1: Channel Discovery and Selection Using a modified snowball sampling approach (Garkava et al. 2024) [18], we systematically identify Telegram channels engaged in identity theft activities through targeted keyword searches for financial fraud and identity theft terminology, followed by network exploration of cross-channel relationships and user migration patterns.

Stage 2: Automated Data Collection We implement continuous monitoring systems to capture the ephemeral nature of communications across identified channels, addressing the temporal characteristics of criminal activities and frequent content deletion practices observed in these environments.

Stage 3: Structured Data Processing and Parsing Raw communications undergo transformation into structured datasets through machine learning-based entity recognition and JSON parsing techniques, enabling systematic categorization of different PII types and preparation for quantitative analysis.

Stage 4: Data Validation and Quality Assessment Collected data undergoes comprehensive validation procedures including geocoding verification, institutional identifier matching, and duplicate detection to ensure analytical reliability and accuracy of geographic and institutional targeting patterns.

Stage 5: Trend Analysis and Pattern Recognition Structured and validated data is subjected to statistical analysis to identify temporal trends, geographic concentrations, institutional targeting patterns, and victimization characteristics that reveal the operational dynamics of identity theft ecosystems on Telegram.

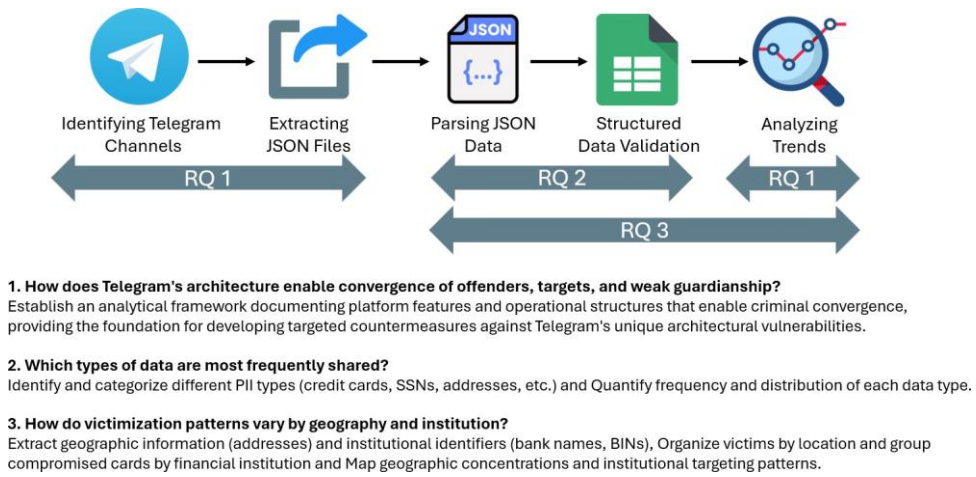

Fig. 3 Mapping Research Questions to Our Methodology

3.2 Understanding Telegram's Platform and Architecture

Telegram is a cloud-based messaging platform launched in 2013 [46] that has evolved from a consumer communication tool into a significant venue for criminal activities. Unlike traditional messaging applications, Telegram combines several unique features that make it particularly attractive to cybercriminals engaged in identity theft operations.

Platform Overview and Accessibility: Telegram operates on the surface web rather than requiring specialized browsers or configurations like traditional darknet markets. This accessibility dramatically expands the potential participant pool beyond the technically sophisticated users typically found on darknet markets. Users can access Telegram through web browsers, mobile applications, or desktop clients, creating multiple entry points for criminal participation.

Communication Architecture: Telegram supports two primary communication formats relevant to criminal operations: channels and groups [47]. Channels are primarily one-to-many broadcast mediums, where only administrators can post content, often used as repositories or marketplaces for stolen data. The audience of subscribers have limited interaction, where they can view content but not respond directly within the channel. Channels maintain complete message history, making them valuable for longitudinal criminal operations. In contrast, groups enable many-to-many interactions and

are commonly used for discussions or negotiations among cybercriminals. Groups may have various permission structures limiting who can post or read messages, creating tiered access models that mirror traditional criminal hierarchies.

Security and Anonymity Features: Telegram offers several features that appeal to criminal users seeking operational security. Its anonymity features permit users to register without providing identifiable credentials beyond a phone number, which can be acquired through burner SIM cards or virtual numbers. The platform also offers secret chats that ensure messages are not stored on servers and self-destructing message capabilities that reduce traceability [48].

Criminal Operational Advantages: Unlike other messaging applications such as Signal and WhatsApp, Telegram allows large public and private groups with thousands of members, creating economies of scale for criminal operations [49]. The platform's channel and group functionality enable criminals to create tiered operations with different levels of access and trust, mirroring organizational structures found in legitimate businesses but with enhanced operational security.

We focused on public channels for their structured content, persistent history, and fewer ethical complexity compared to monitoring interpersonal communications.

Note on Encryption: While Telegram provides end-to-end encryption for private chats and secret conversations, this study focuses exclusively on public channels where content is openly accessible without encryption. Our research examines how Telegram's public channel infrastructure facilitates criminal activities, not its encrypted communication features.

3.3 Criminal Convergence and Supply Chain Transformation

The combination of Telegram's architectural features creates what we term a "Transformative Criminal Convergence Space", reshaping how identity theft operations function compared to traditional criminal supply chains. This transformation requires understanding both the criminal methods employed and how traditional supply chain models have evolved on this platform.

Criminal Methods and Operational Tactics: Within Telegram's criminal ecosystems, cybercriminals employ various tactics to acquire and distribute stolen personal and financial information. Digital methods predominate, with phishing campaigns specifically designed to gather credentials that can be immediately monetized. These attacks often disguise themselves as quiz links, free giveaways, or surveys that trick individuals into revealing their credentials or unknowingly installing malware. Smishing, a variation of phishing that leverages text messages, capitalizes on the extensive use of mobile phones by sending malicious links via SMS. Social engineering tactics are refined within these communities, with criminals sharing successful impersonation strategies and templates for trusted entities.

The transition to Telegram has also facilitated hybrid approaches combining digital and physical methods. Information obtained through shoulder surfing, discreetly

observing potential victims in public spaces, can be immediately shared with criminal networks for rapid exploitation. Insider threats have evolved, with Telegram channels sometimes offering bounties for employees willing to extract confidential data from financial institutions. Traditional physical methods like mailbox theft and wallet theft remain relevant, with criminals using Telegram channels to coordinate geographic targeting based on resource indicators and security vulnerabilities.

Reconfigured Supply Chain Structure: Understanding these criminal methods requires examining how they fit within the reconfigured Telegram-based supply chain (Fig. 4). In this new model, suppliers include both specialized data thieves who focus on specific acquisition methods and opportunistic contributors who share credentials obtained through various means. Manufacturers develop tools and templates optimized for collecting credentials through Telegram-based distribution, creating more accessible solutions than the sophisticated technical infrastructure typically required for darknet operations. Distributors leverage Telegram's broadcast functionality to rapidly disperse stolen information across multiple channels simultaneously, creating redundancy that makes comprehensive takedowns nearly impossible. Retailers operate within these channels by offering curated selections of high-value credentials, often with real-time verification services to ensure validity at the time of purchase. This role fluidity is represented by the bidirectional arrows in the supply chain model, indicating that participants can simultaneously occupy multiple roles within the ecosystem, for example, a distributor broadcasting stolen credentials through channels may also act as a supplier by contributing newly acquired data, while retailers offering curated selections can simultaneously function as manufacturers by developing verification tools or credential templates.

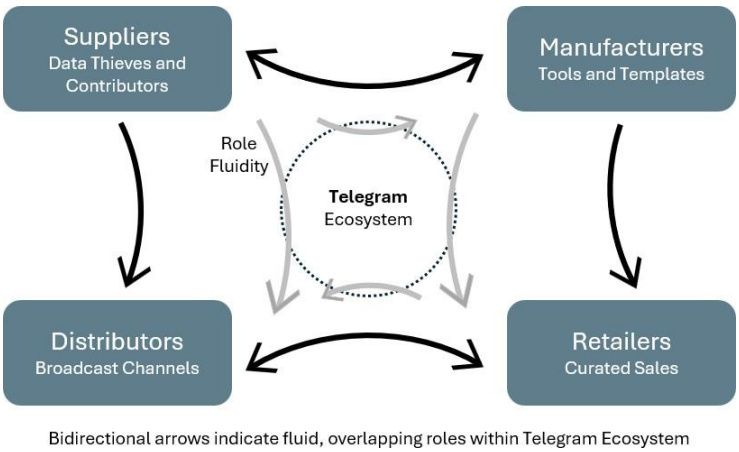

Fig. 4 Telegram-Based Supply Chain

In contrast to the traditional underground supply chain (Fig. 2), Telegram-based supply chain model benefits from Telegram's combination of accessibility and anonymity features, allowing participants to enter and exit the ecosystem with minimal technical barriers. The ease of channel creation and termination enables rapid adaptation to detection efforts, where operations simply migrate to new channels when facing scrutiny. This process has created a shadow economy that mirrors legitimate business practices but operates with greater agility and resilience to disruption than traditional darknet marketplaces.

Criminal Convergence Theory Application: These features collectively transform Telegram into an environment where the three elements of Routine Activity Theory (motivated offenders, suitable targets, and lack of capable guardianship) interact in unique ways. This platform lowers barriers for motivated offenders through its accessibility and user-friendly interface, centralizes access to suitable targets through its broadcast functionality and large group capabilities, and creates obstacles to effective guardianship through its technical features and operational characteristics. Telegram enhances these operations by providing immediate feedback on which techniques yield the most valuable data, creating a learning ecosystem where methods are continuously refined and adapted to changing security measures.

3.4 Channel Sampling Strategy and Selection Criteria

The selection of appropriate Telegram channels is necessary to ensure that our dataset accurately represents the identity theft ecosystem while maintaining ethical and methodological rigor. Channel selection follows a structured approach to ensure consistency and relevance. Selection criteria for inclusion in our study require channels to have: (1) at least 50 active participants engaged in identity theft discussions, (2) publicly accessible chat history, (3) explicit references to financial credential trading, and (4) a public URL allowing researcher access without invitation barriers. These criteria align with the methodological framework established by Wang et al. (2024) [19] in their large-scale analysis of cybercriminal channels on Telegram.

The modified snowball sampling approach begins with seed channels identified through targeted keyword searches for identity theft terminology, followed by network exploration of cross-channel relationships and user migration patterns. Sample expansion occurs through systematic tracing of cross-channel links, tracking user migrations between spaces, and observing channel recommendations. This comprehensive approach ensures diverse representation of the identity theft ecosystem while maintaining systematic consistency.

3.5 Data Categories and Analytical Framework

Our analysis targets two primary data categories that constitute the foundation of identity theft operations on Telegram. Textual data (Fig. 5) encompasses the majority of our analytical focus, including financial credentials (credit card numbers, expiration

dates, CVV codes, banking details), personal identifiers (full names, home addresses, phone numbers, dates of birth), account access information (usernames, passwords, associated email addresses), government-issued identification numbers (social security numbers), and various other types of sensitive personal data.

Fig. 5 Sample Textual Data Posted on Telegram

Media data (Fig. 6), while not the primary focus of our analysis, provides crucial contextual evidence including images of identification documents, photographs of physical credit cards, screenshots demonstrating account access, and occasionally instructional videos detailing fraud techniques. These media elements often accompany textual data as verification of authenticity or demonstrations of access methods.

Fig. 6 Sample Media Files Posted on Telegram

Our analytical framework employs three integrated dimensions to examine identity theft patterns and address our research questions:

- *Typological Analysis* categorizes and quantifies different types of PII appearing in Telegram channels to identify which forms of personal information are most trafficked. This analysis reveals criminal preferences within these ecosystems and provides insight into monetization strategies by examining the prevalence of specific data types.
- *Spatial-Temporal Analysis* examines how stolen financial information is distributed across geographic regions and time periods. Geographic analysis based on address data and BIN information reveals targeting patterns, while temporal analysis identifies cyclical patterns and strategic timing in criminal activities to understand operational cycles.
- *Institutional Analysis* analyses which financial institutions and payment systems appear most frequently in compromised credentials. This dimension provides insight

into targeting strategies and potential institutional vulnerabilities within the financial ecosystem by detecting patterns across different banks, card types, and geographic regions.

These integrated analytical dimensions develop a comprehensive understanding of identity theft operations on Telegram channels, addressing questions about how stolen information is organized, what types of information are prioritized, and what distribution patterns emerge in social messaging environments.

4 Implementation

The implementation of our research methodology required developing novel technical approaches to overcome the unique challenges posed by Telegram's messaging environment. Unlike traditional web scraping or Application Programming Interface (API)-based data collection methods commonly used in cybercrime research, studying identity theft on Telegram demanded specialized tools capable of navigating dynamic, constantly changing criminal ecosystems while respecting platform constraints and ethical boundaries. This section details our technical implementation, including the automated data collection framework, channel identification protocols, and data processing pipelines that enabled systematic analysis of identity theft activities across 25 Telegram channels throughout 2023. Complete source files for the custom text parsing scripts, data extraction tools, and analytical frameworks developed for this research are accessible through the code repository referenced in Appendix D.

4.1 Channel Identification and Network Mapping Implementation

Initial seed channels were selected based on activity level, public accessibility, and relevance to identity theft operations. The implementation of our modified snowball sampling approach resulted in the identification of 25 high-value channels with sustained activity related to identity theft. These channels varied significantly in size, specialization, and operational practices, providing a diverse representation of the identity theft ecosystem on Telegram.

Keyword Search Implementation: The seed channel identification process utilized targeted keyword searches for identity theft jargon including "fullz" (complete identity packages), "dumps"(raw credit card data), "logs"(compromised account credentials), and "drops"(locations for receiving fraudulent goods). Additional terminology includes "checks", "accounts", "bank names", and fraudster slang such as "glass" (counterfeit money), "slips" (fraudulent checks), "wells/welly"(Wells Fargo bank accounts used for illegal purposes), "navy" (Navy Federal Credit Union), and "chase" (JPMorgan Chase bank). This comprehensive keyword strategy ensured broad coverage of different criminal specializations and operational approaches.

The network expansion process was implemented through systematic documentation of cross-channel links, user migration tracking between spaces, and channel recommendation analysis. This technical implementation enabled comprehensive mapping of the interconnected nature of criminal networks operating across multiple Telegram channels.

4.2 Automated Data Collection System Implementation

To manage the volume and ephemeral nature of communications on these channels, we implemented an automated collection system designed specifically for Telegram's interface. This system operated continuously throughout 2023, providing a comprehensive longitudinal dataset of identity theft activities.

Technical Architecture and Tools: Our automated monitoring system operated through Sikuli IDE (Yeh et al., 2009), an open-source automation tool designed to interact with graphical user interfaces using image recognition (Figure 4.5). This application interacted with Telegram's web interface rather than its API, which provided limited access to public channel data. This approach allowed us to capture content that would otherwise be inaccessible while maintaining ethical boundaries by only collecting publicly available information. Sikuli allowed us to recognize on-screen elements such as buttons, icons, and text fields, and simulate mouse clicks and keyboard inputs accordingly.

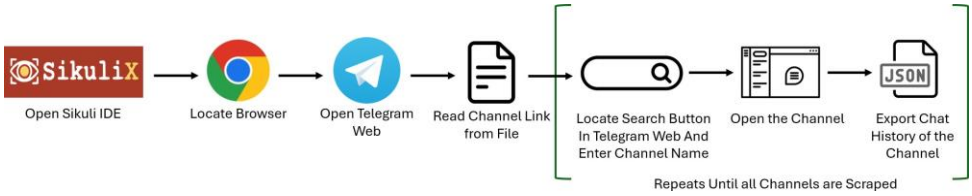

Fig. 7 Data Scraping using Sikuli IDE - Overview

Operational Workflow and Automation: The Sikuli-based scraper was configured to perform a series of automated tasks: launching a web browser, accessing Telegram Web, searching for and entering identified channels, navigating through content, and exporting chat logs as JavaScript Object Notation (JSON) files to our storage repository (See Appendix E for a detailed scrapping flowchart). This automation was scheduled to run weekly, ensuring that newly posted content was captured and changes to older messages were recorded. Similar methodological approaches have been validated in previous studies of criminal activities on messaging platforms.

Data Capture and Content Management : The monitoring process followed a structured workflow beginning with channel authentication, followed by systematic content extraction from publicly visible posts. Content was captured on a regular basis to

ensure comprehensive coverage, including both new posts and updates to existing content. This frequent collection schedule addressed the common practice of content deletion or modification observed in criminal channels, ensuring that ephemeral information was properly documented.

4.3 Structured Data Extraction Implementation

The final stage of our data collection framework involved transforming the raw monitored content into structured, analyzable data. This process was essential for enabling systematic analysis of identity theft patterns across different channels and time periods.

The following implementation steps detail our data extraction and processing pipeline designed to convert raw Telegram communications into structured, analyzable datasets:

Primary Processing Architecture: Raw chat logs stored in JSON format typically contained unstructured text cluttered with advertisements, user aliases, timestamps, and platform-specific metadata. To convert this unstructured data into well-organized records of PII, we employed a Python-based Amazon Web Services (AWS) Comprehend [50] JSON Parser and Excel Export Tool, which integrated machine learning-based entity recognition from AWS Comprehend with additional APIs for geocoding and data enrichment.

Media data was organized based on content type, associated textual information, and apparent purpose within identity theft operations. While detailed analysis of media content fell outside the scope of this study, this cataloging approach provided valuable context for understanding the broader ecosystem.

Automated Text Processing Pipeline: The text parser automated several essential tasks that would be prohibitively time-consuming if performed manually. It began by scanning directories for JSON files containing Telegram channel logs, identifying relevant text fields within complex nested structures. Once relevant content was isolated, the parser cleaned and pre-processed it to remove duplicates, invalid characters, and irrelevant metadata that could interfere with entity recognition.

Machine Learning-Based Entity Recognition: At the core of our extraction process was AWS Comprehend, a Natural Language Processing (NLP) service that uses machine learning to identify entities within text. We utilized two primary API calls: `detect_entities()` to find general named entities like people, organizations, and locations, and `detect_pii_entities()` to specifically target sensitive information such as credit card numbers and social security numbers. This dual approach provided comprehensive coverage of both general contextual information and specific PII.

Financial Data Validation and Enhancement : For financial data enrichment, we integrated the BINChecker API [51] to validate and categorize credit card information found within the dataset. The BIN (Bank Identification Number) checking

process enabled identification of issuing banks, card types (Visa, Mastercard, American Express), and geographic regions associated with compromised payment cards. This API integration provided crucial intelligence for institutional analysis and helped verify the authenticity of financial credentials found in criminal channels, distinguishing between legitimate stolen data and fraudulent or test numbers commonly used in scammer communications.

Geographic Data Processing: Once entities were identified, the parser categorized them into predefined fields based on their type and context. This categorization allowed for structured analysis of different PII types, enabling us to identify patterns in the kinds of information being traded. For address data, we implemented additional refinement through a geocoding API [52] that standardized formatting and extracted supplementary details like ZIP codes and states.

Data Export and Structure: The parsed results were exported to Excel format, organizing data into human-readable spreadsheets that maintained connections to source channels and timestamps. This structured format facilitated both quantitative analysis through statistical methods and qualitative assessment through content review. By preserving these connections to original sources, we could trace the provenance of specific data elements when deeper contextual understanding was required.

Post-Processing Workflow Implementation:Following the initial parsing phase, we implemented a comprehensive post-processing workflow to further refine and clean the extracted data. This multi-stage process included file merging operations to consolidate data from multiple channels, address extraction and validation procedures, geocoding enhancement for incomplete location data, and ZIP code filtering to retain only records with sufficient geographic specificity for analysis. These sequential refinement steps ensured that our final dataset contained only the most relevant and complete records for statistical analysis.

Security Protocols and Quality Assurance: Throughout this extraction process, we maintained strict data security protocols. We implemented access controls to ensure that sensitive information remained secure, and all team members followed established protocols for data handling to prevent potential exposure.

The parser included robust error handling for cases such as empty files, invalid JSON structures, or API failures. When encountered, these exceptions were logged and flagged for manual review, ensuring that data quality issues did not silently corrupt our analytical results. This approach maintained the integrity of our dataset even when processing challenging content.

Analytical Capabilities and Output : The resulting structured dataset enabled multidimensional analysis of identity theft patterns on Telegram, allowing us to examine the prevalence of different PII types, track temporal changes in criminal activities, identify geographic concentration of victims, and detect institutional targeting patterns.

This data extraction framework successfully transformed raw communications from 25 Telegram channels into structured datasets, establishing a replicable methodology for analyzing criminal activities on messaging platforms while maintaining data integrity and security throughout the process.

4.4 Accuracy Assessment of PII Extraction and Data Quality Assurance

A critical component of our methodology was validating the reliability of our PII extraction system prior to its application on Telegram data. This evaluation was crucial to establish confidence in our automated analysis capabilities.

Our parser, built around AWS Comprehend's entity recognition capabilities, required systematic validation to ensure its ability to accurately identify and categorize PII under conditions that mirror the inherent messiness of criminal communications. To accomplish this, we developed a three-stage validation framework that progressively tested the system's resilience to noise and obfuscation commonly found in criminal communications, including formatting variations, deliberate character substitutions, and fragmented information designed to evade automated detection.

We established a ground truth dataset containing 2,500 synthetic PII entries that mimicked the patterns and formatting observed in actual Telegram channels during preliminary observation. This dataset was carefully constructed to represent the full spectrum of PII types encountered in identity theft operations while avoiding privacy concerns associated with using real compromised data.

We created three increasingly challenging test conditions by introducing controlled noise to the ground truth data:

- *Stage 1 (Low Noise):* Minor formatting variations and simple obfuscation techniques.
- *Stage 2 (Medium Noise):* Fragmented information and moderate obfuscation patterns.
- *Stage 3 (High Noise):* Complex obfuscation, significant formatting disruptions, and deliberate misdirection.

This approach allowed us to simulate the variety of data quality conditions encountered in actual Telegram channels, from relatively clean credential dumps to deliberately obfuscated communications.

We then executed our parser against each dataset, applying normalization procedures to maximize pattern recognition while preserving semantic integrity. All text was lowercased and stripped of extra symbols before comparison, with accuracy evaluated through exact match checks and substring detection. Results were evaluated using a multi-dimensional accuracy framework that considered exact matches, partial matches, and false positives.

Our assessment revealed that the parser maintained approximately 72% accuracy even under high-noise conditions, with perfect accuracy achieved on clean data (Fig. 8). This performance level provides sufficient reliability for identifying meaningful patterns in Telegram communications while acknowledging inherent limitations when analyzing deliberately obfuscated content.

The validation confirmed that our methodological approach was robust enough to extract valuable insights from the noisy, inconsistent environment of Telegram channels. This level of accuracy provides a transparent foundation for interpreting the findings presented in subsequent sections.

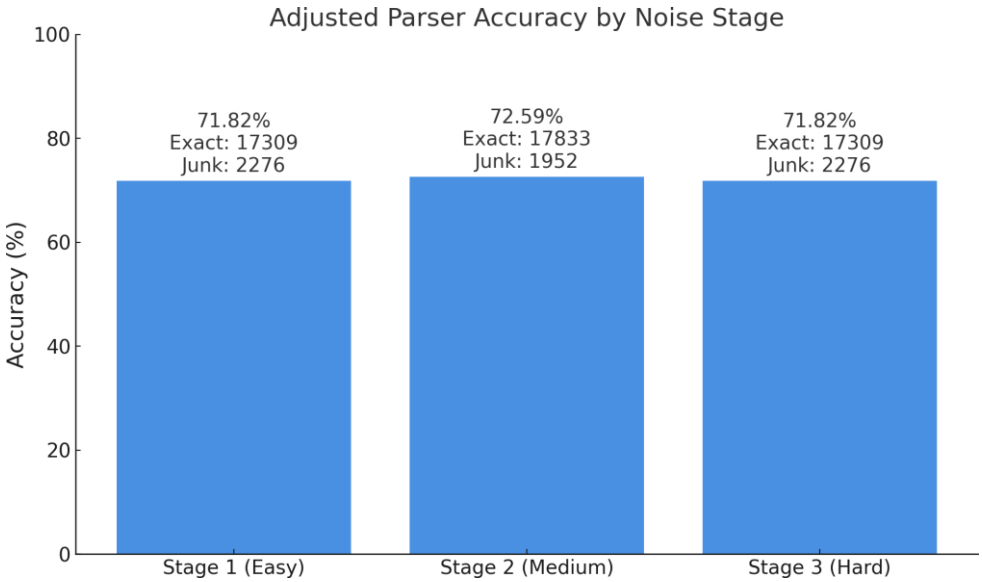

Fig. 8 Adjusted Parser Accuracy Results

Other Data Validation and Quality Checks: Our methodology incorporated several other validation mechanisms to strengthen the credibility of our findings. This multi-layered approach to validation ensured that our findings were based on reliable data despite the challenging nature of the research environment.

We implemented cross-channel verification by examining information that appeared across multiple channels to identify duplicates and confirm consistency in format and content. This process revealed patterns in how stolen data propagates across criminal networks and provided confidence in the authenticity of observed credentials. By tracking the movement of specific data packages between channels, we could better understand the interconnected nature of these criminal ecosystems.

Financial credential validation formed another key component of our quality assurance process. Credit card numbers were validated using industry-standard verification algorithms such as the Luhn algorithm to distinguish potentially valid card numbers from randomly generated sequences or deliberately falsified information. Bank Identification Numbers (BIN) were cross-checked against established databases to confirm the plausibility of issuing institutions and card types, providing an additional layer of verification for our financial data analysis.

Address data underwent geocoding to verify that locations corresponded to actual physical addresses. This process revealed that only approximately 25% of addresses could be successfully geocoded, highlighting significant data quality issues within criminal channels. This finding provides important context for interpreting geographic distribution patterns and suggests deliberate obfuscation tactics or careless data handling by channel operators. The geocoded addresses that were successfully validated provided valuable insights into potential geographic targeting patterns.

All scraped chat logs were uploaded to a Dropbox repository for centralized storage and version control. This approach enabled multiple team members or automated processes to access the data consistently. By maintaining a folder structure based on channels and scraping dates, researchers could track the growth of each channel's conversation history, compare snapshots of data across different time frames, and recover earlier files in the event of data corruption or errors in subsequent processing steps.

5 Analysis and Results

With this robust technical infrastructure and analytical capability in place, beginning with how Telegram's architecture enables the convergence of offenders, targets, and weak guardianship in new ways (RQ1), followed by an examination of which types of PII are most frequently shared in these criminal ecosystems (RQ2) and concluding with analysis of how victimization patterns vary by geography and institution (RQ3).

5.1 How Telegram's Architecture Enables Criminal Convergence? (RQ1)

Telegram's architecture fundamentally transforms the convergence of Routine Activity Theory's three criminal elements by creating a "Transformative Criminal Convergence Space" that operates distinctly from traditional cybercrime environments. Our comprehensive analysis of 25 channels throughout 2023, utilizing automated data collection systems and machine learning-based entity recognition, reveals how Telegram's unique combination of surface web accessibility, minimal encryption barriers in public channels, and scalable communication structures eliminates traditional barriers while maintaining operational security. This analysis demonstrates how the platform's architectural features systematically enable criminal convergence across three key dimensions.

Expanding Motivated Offenders: Our framework identifies how the platform's channel accessibility documentation shows that unlike darknet markets requiring technical expertise (creating guardianship through complexity), Telegram's user-friendly interface dramatically expands the pool of motivated offenders by removing technical barriers to entry, allowing anyone with basic smartphone skills to participate in identity theft operations that generated over 400,000 PII entries in our dataset alone. This finding demonstrates that architectural accessibility directly correlates with criminal

participation rates, indicating that intervention strategies must target accessibility reduction without compromising legitimate privacy needs.

Centralizing Suitable Targets: The framework demonstrates how the platform's channel and group architecture enable unprecedented access to suitable targets through broadcast functionality that can instantly distribute stolen credentials to thousands of subscribers. Our empirical findings show this convergence operating at massive scale, with credit card data comprising 82% of all shared PII, demonstrating the platform's effectiveness in centralizing high-value targets for criminal exploitation. The hybrid public-private structure allows criminals to create tiered operations with different access levels that mirror legitimate business hierarchies, evidenced by our observation of specialized channels for different types of financial credentials. These findings indicate that target protection mechanisms must account for large-scale, simultaneous distribution capabilities.

Weakening Capable Guardianship: Our framework reveals systematic weaknesses in capable guardianship created by Telegram's anonymity features, self-destructing messages, and minimal registration requirements that impede traditional law enforcement monitoring techniques. Our automated data collection implementation demonstrated how the platform's technical architecture facilitates persistent criminal ecosystems where participants operate across multiple channels simultaneously, with cross-channel networking creating redundancy that makes comprehensive takedowns nearly impossible. This finding suggests that enhanced guardianship mechanisms must be designed to operate within distributed, resilient criminal networks while respecting privacy considerations.

Addressing Architectural Challenges: The operational dynamics identified through our framework demonstrate how platform architecture facilitates criminal activity through immediate feedback mechanisms that enable rapid adaptation to security measures, the ease of channel creation and termination that allows operations to migrate when facing scrutiny, and the transformation from linear darknet supply chains to fluid, overlapping criminal roles within Telegram's ecosystem. Our implementation findings reveal that the platform's web interface accessibility, combined with its broadcast functionality and persistent message history in channels, creates an environment where criminal convergence occurs continuously rather than requiring specific temporal or spatial alignment, fundamentally altering the traditional crime opportunity structure.

This architectural analysis provides the analytical foundation for developing effective countermeasures that address the platform's specific characteristics. The framework enables the design of real-time monitoring systems that can adapt to rapid channel migration, multi-channel tracking capabilities that account for the distributed nature of operations, and intervention strategies that leverage the platform's centralized infrastructure while respecting privacy considerations. Without this systematic framework for understanding how Telegram's architecture enables criminal convergence, traditional cybersecurity approaches remain inadequate for addressing these challenges.

5.2 Which Types of Data Are Most Frequently Shared? (RQ2)

Understanding which types of PII are most frequently trafficked on Telegram provides crucial insights into criminal preferences and monetization strategies within identity theft networks. We examine the distribution patterns of different PII categories, temporal variations in data sharing, and victim demographic characteristics to reveal the operational priorities and market dynamics of criminal ecosystems operating on the platform.

Overall Distribution

Our analysis of the data collected from Telegram channels exposes distinct patterns in the types of personal information being trafficked. As depicted in Fig. 9, financial credentials heavily dominate the illicit marketplace, with credit card numbers accounting for 82% of the total PII collected. This frequent occurrence of credit card information likely stems from its immediate financial utility to cybercriminals and the established infrastructure for exploiting this data through automated transactions, making it a more attractive target than other forms of personal information. This dominance supports the economic motivations inherent in identity crime and aligns with low-risk/high-return criminological models discussed in theoretical frameworks.

Card Verification Values (CVV) represent 10% of the compromised data, demonstrating their significance as essential security components that enable fraudulent online purchases where physical cards are not required. Physical addresses account for 3% of the data, serving as important components for identity verification and physical mail fraud schemes. Victim names and phone numbers each constitute 2%, while dates of birth make up 1%. Social Security Numbers represent the smallest fraction, comprising just 0.01% of the total dataset.

The extremely low occurrence of SSNs could suggest that despite their significant value for identity theft, they are either more securely protected, less frequently breached, or traded in more exclusive criminal venues rather than Telegram's semi-public channels. This finding indicates that while Telegram serves as an accessible platform for basic financial fraud, more sophisticated identity theft operations requiring government-issued identifiers may operate through different channels or require higher levels of trust and exclusivity. This specialization pattern provides important insights into how different types of PII are valued and distributed within Telegram's identity theft ecosystem.

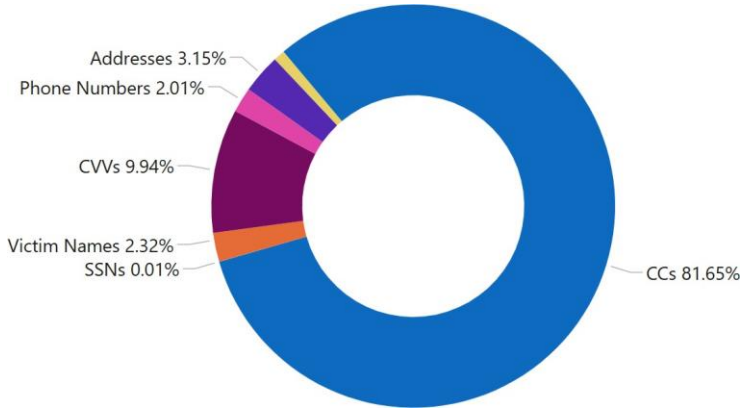

Fig. 9 Percentage Distribution of PII Categories Collected in 2023

Temporal Patterns in Compromised Financial Data

Our longitudinal analysis reveals significant monthly variations in the volume of compromised financial credentials being circulated, potentially indicating seasonal patterns in criminal activity, data breach events, or fluctuations in market demand. Fig. 10 illustrates the monthly distribution of credit card numbers, CVV entries, and expiry details throughout 2023.

Credit card numbers showed distinct volume peaks in June (76,000), January (70,000), and March (58,000). In contrast, November and December exhibit minimal activity with the lowest counts, each accounting for approximately 1,000 entries. This pattern suggests potential correlation with major shopping seasons when increased card usage creates more opportunities for compromise through skimming, phishing, or point-of-sale breaches. The timing of peaks aligns with post-holiday shopping in January, spring break travel in March, and summer vacation periods in June, supporting the hypothesis that criminals exploit periods of heightened consumer activity.

The dramatic decline in November and December is counter-intuitive given the holiday shopping season, potentially indicating either a lag between compromise and marketplace appearance or a shift in criminal tactics during high-alert periods when financial institutions and consumers exercise greater vigilance. This temporal pattern provides valuable intelligence for financial institutions seeking to optimize their fraud detection systems and for law enforcement agencies planning resource allocation.

CVV entries mirrored this variability but with different peak months, with March (15,351) and April (11,360) marking the highest volumes, then dropping dramatically in July (157). The inconsistent pattern between card numbers and CVVs suggests that these elements may be sourced from different breach tactics or represent different stages of criminal processing. Similarly, expiry details fluctuated monthly, with June

showing the highest frequency (1,428), followed by March (1,374) and May (1,253), with December registering the minimal count (221). The alignment of expiry date peaks with card number peaks in June suggests coordinated releases of complete card data packages, potentially indicating major breaches that captured full card details rather than partial information.

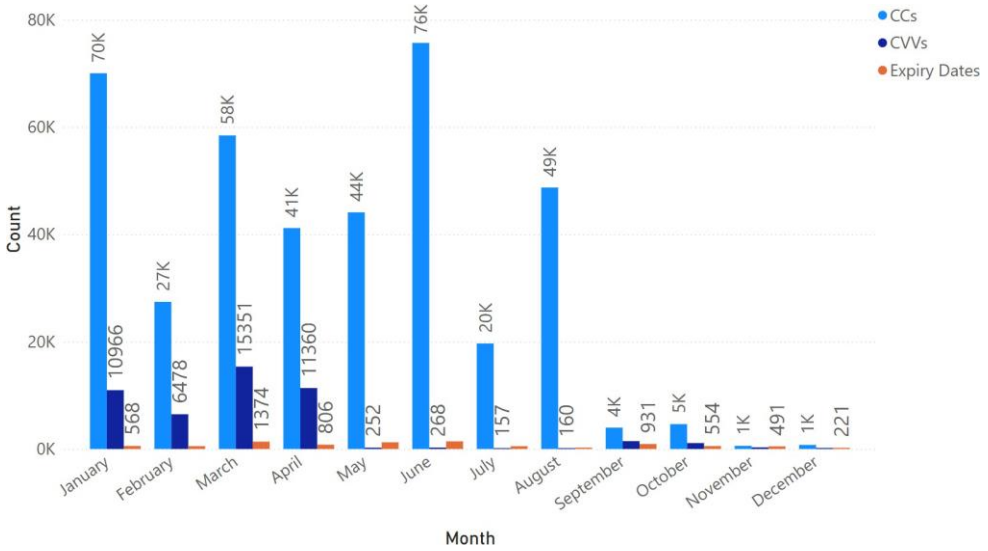

Fig. 10 Total Compromised Credit Cards Collected in 2023

Victim Identity Information Analysis

Beyond financial credentials, our research captured significant volumes of PII that provide insight into victim demographics and geographic distribution. Fig. 11 outlines the monthly collection trends for four categories of personal data: victim names, dates of birth, phone numbers, and social security numbers.

Victim names were most frequently recorded in September (2,066), followed by October (1,872), March (1,496), and November, which registered the lowest volume (318). Date of birth collections peaked in September (545), followed closely by January (544) and May (459), while December saw the minimal count (111). Phone numbers exhibited the most dramatic monthly variation among identity elements, with a pronounced spike in March (2,215), followed by secondary peaks in June (1,715) and February (1,426), and a stark decline to November's annual low (75). Social Security Numbers consistently demonstrated the lowest collection volumes throughout the year, never exceeding 13 instances per month, with the highest count occurring in October.

The correlation between different PII types across specific months, particularly September and October, suggests coordinated releases of data following major breach

events that captured complete identity records rather than isolated data points. These synchronized patterns can provide security researchers with potential timestamps for investigating major breach events that may not yet have been publicly disclosed. The consistency of these patterns across multiple data types strengthens confidence in their authenticity and suggests that major data breaches tend to surface in criminal marketplaces with predictable timing.

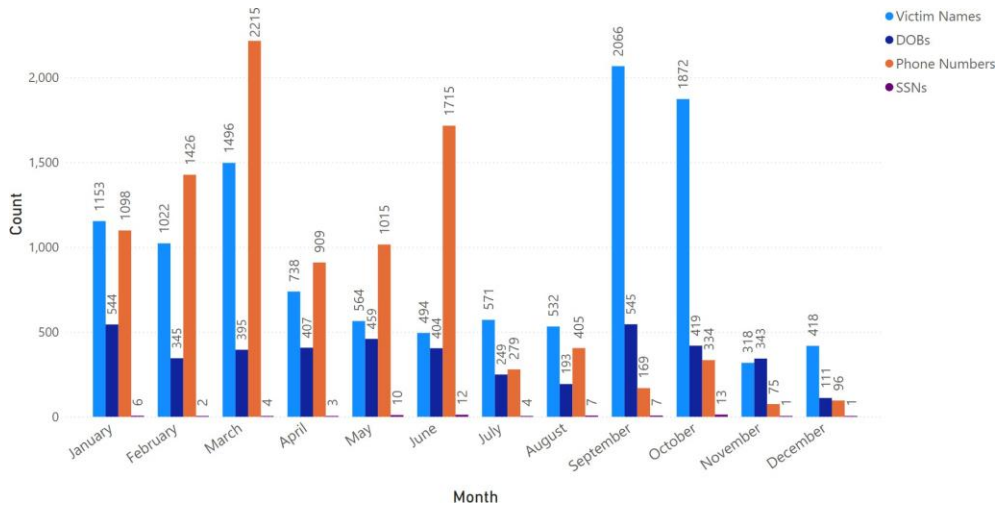

Fig. 11 Compromised Victim Information Collected in 2023

5.3 How Do Victimization Patterns Vary by Geography and Institution? (RQ3)

Understanding how victimization patterns vary across geographic regions and financial institutions reveals critical insights into criminal targeting strategies and the systematic nature of identity theft operations on Telegram. We examine the geographic distribution of compromised victims, analyze targeting patterns across different financial institutions, and investigate payment network vulnerabilities to identify the strategic priorities and operational intelligence that drive criminal decision-making within these ecosystems.

Geographic Distribution Analysis

The geographic analysis of compromised addresses provides critical insights into the regional distribution of identity theft victims and potential targeting patterns by cybercriminals. From a total of 15,240 addresses collected throughout the year, significant monthly variations were observed (Fig. 12), with peaks in September (2,463) and October (2,420), aligning with the patterns observed for other PII types. The lowest counts were recorded in December (488) and November (325).

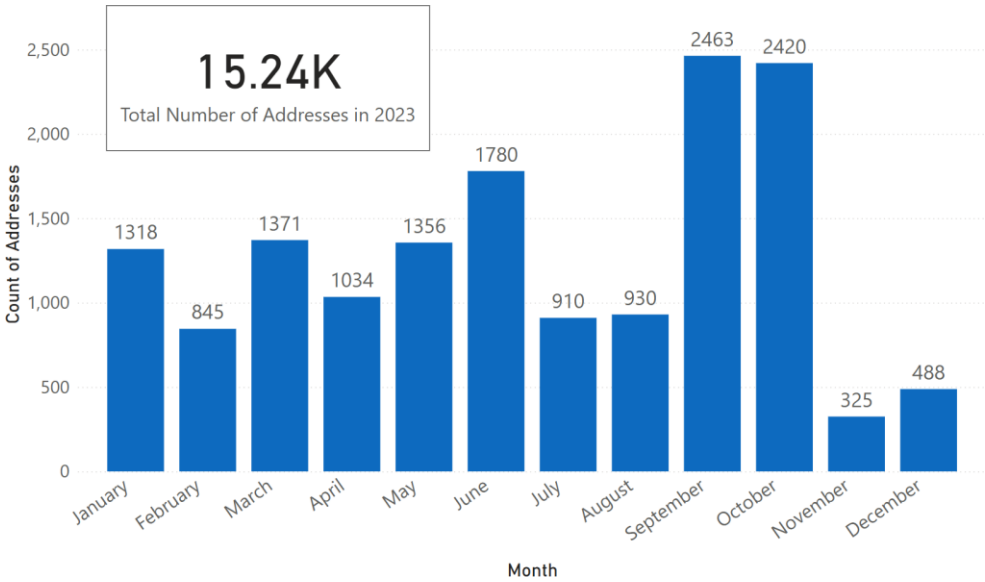

Fig. 12 Total Compromised Victim Addresses Collected in 2023

An important finding emerged during address validation: only 25% (3,720) of the raw address data yielded valid, geocodable addresses (Fig. 13). This substantial verification failure rate highlights significant data quality issues within criminal channels, including deliberately falsified information, formatting inconsistencies, incomplete entries, and intentional obfuscation to avoid detection. The geocoded addresses maintained similar temporal distribution patterns as the raw data, with highest valid address counts in September (946) and October (841), and lowest in May (134) and November (122). This consistency suggests that quality issues affect all months proportionally rather than being concentrated in specific time periods, indicating systematic data quality problems across the ecosystem.

Fig. 13 Total Compromised Geocoded Victim Addresses Collected in 2023

The state-level analysis of geocoded addresses (Fig. 14) reveals a distinct geographic pattern that supports our hypothesis regarding correlation with population centers. The graph displays the total number of identified zip codes within each state, with the total number of addresses shown in parentheses. Texas led with the highest number of victims (301 addresses), followed closely by California (224 addresses) and Florida (210 addresses). Other significantly affected states included New York (182 addresses), Pennsylvania (133 addresses), and Illinois (109 addresses). In contrast, less populated states such as Wyoming (5 addresses) and Vermont (4 addresses) showed minimal representation in the dataset.

This distribution broadly mirrors population density patterns across the United States, with the top six states by victim count ranking among the ten most populous states. However, the geographic patterns also suggest factors beyond simple population correlation. Texas shows the highest representation despite having 8.7% of the U.S. population, while California appears underrepresented despite comprising 11.9% of the population. This suggests that factors beyond population size could influence victimization rates, including regional differences in cybersecurity awareness, consumer behaviors or targeted attacks on region-specific institutions.

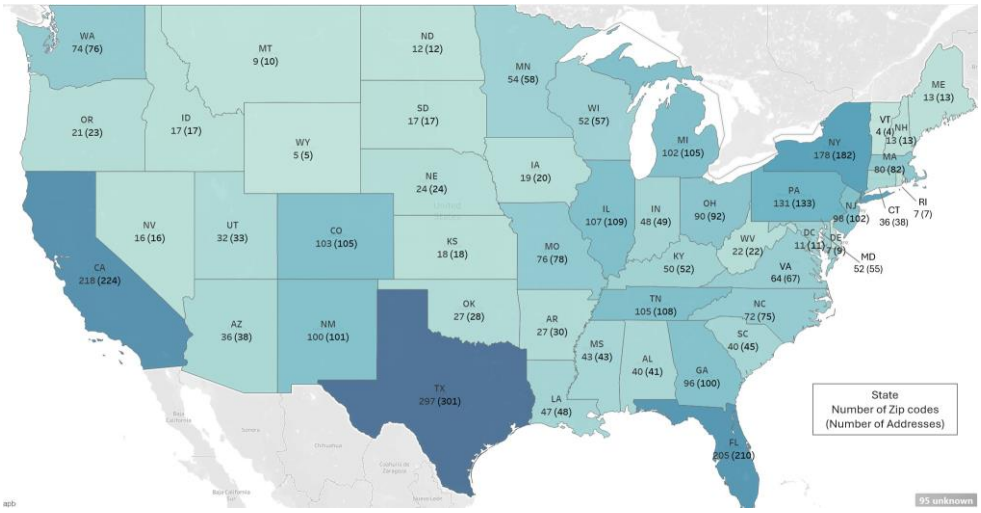

Fig. 14 Geographic Distribution of Geocoded Compromised Victim Addresses in 2023

Financial Institution Analysis

The analysis of compromised cards by financial institution provides valuable intelligence on which organizations face the greatest threats from identity theft and reveals potential patterns in criminal targeting strategies. Our dataset captured compromised credentials from top 100 distinct financial institutions, operating in the United States, based on total consolidated assets as reported to the Federal Reserve System. To protect institutional reputations and maintain research ethics, all banks have been anonymized using generic identifiers (Bank A, Bank B, etc.), with the complete list provided in Appendix C. This anonymization approach allows us to analyze systemic targeting patterns while avoiding potential reputational harm to specific institutions.

Fig. 15 shows that Bank A experienced the highest volume of compromised cards (12,027), followed by Bank B (9,030). The subsequent institutions, from Bank C through Bank J, show a gradual decline in their card counts. The distribution reflects a clear concentration among major financial institutions, with the top two banks accounting for 33.8% of all compromised cards with identifiable issuers in our dataset. Bank A, with consolidated assets of $3.87 trillion, shows the highest compromise rate, consistent with its position as one of the largest U.S. financial institutions. This pattern aligns with our hypothesis that larger financial institutions with extensive customer bases are more aggressively targeted by fraudsters, leading to higher volumes of compromised accounts.

The targeting pattern suggests that criminals prioritize larger banks due to several strategic advantages. Broader customer bases create larger potential victim pools, well-known institutions may yield higher success rates in social engineering attacks due to greater public familiarity, and standardized security practices at major banks may allow criminals to scale attacks across multiple victims using similar techniques. This

concentration of criminal attention on major institutions has important implications for both the targeted banks and the broader financial ecosystem.

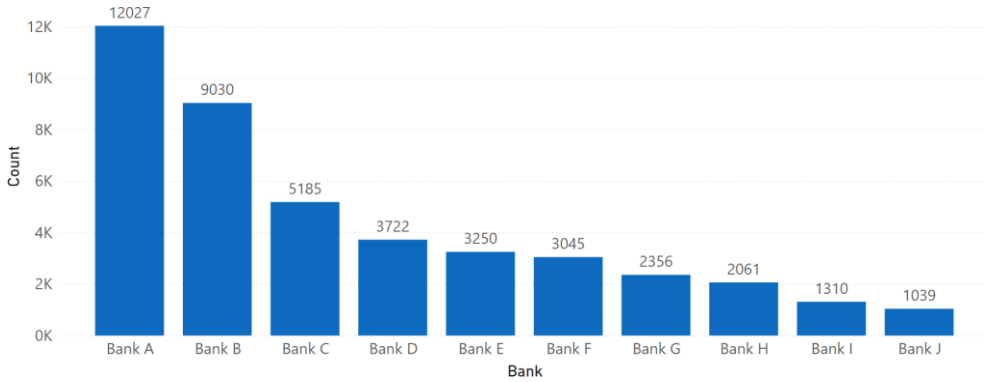

Fig. 15 Top 10 Compromised Banks by Count in 2023
* Bank Brand Consolidated Assets (Mil $) 2023: Bank A = 3,875,393; Bank B = 271,461; Bank C = 3,180,000; Bank D = 1,060,978; Bank E = 1,932,468; Bank F = 1,477,487; Bank G = 478,464; Bank H = 663,491; Bank I = 389,865; Bank J = 77,176

Further analysis of compromised accounts by card issuing organizations (Fig. 16) shows that Organization A dominates with the highest number of breaches (13,111), followed by Organization B (7,175) and Organization C (6,709). A steady decline follows across Organizations D through J, with Organization I (1,694) and Organization J (1,252) recording the lowest figures. This distribution pattern reinforces the finding that criminal activities concentrate on the largest and most recognizable financial institutions.

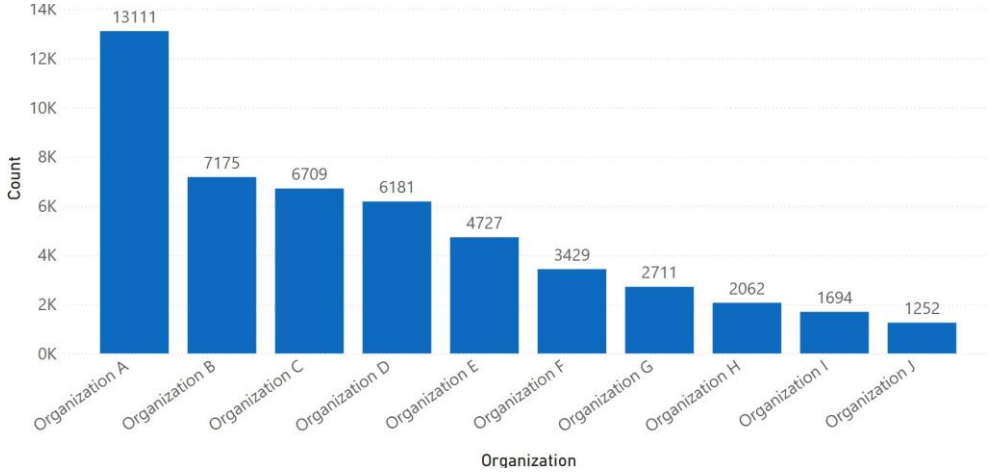

Fig. 16 Top 10 Compromised Financial Organizations by Count in 2023
* Organization's Consolidated Assets (Mil $) 2023: Organization A = 3,875,390; Organization B = 3,261,520; Organization C = 78, 500; Organization D = 3,875,390; Organization E = 1,930,000; Organization F = 3,261,520; Organization G = 170,801,100; Organization H = Unknown; Organization I = 478, 500; Organization J = 663,000

The distribution across payment networks (Fig. 17) shows that Visa (93,416 cards) and Mastercard (89,449 cards) account for most compromised cards, with American Express ranking third (13,276 cards). Smaller networks like Discover (1,135 cards) and regional payment schemes show substantially lower compromise volumes. This pattern closely aligns with the worldwide distribution of these payment networks in the consumer market, suggesting that criminal activity is proportional to card prevalence among consumers rather than targeting specific network vulnerabilities.

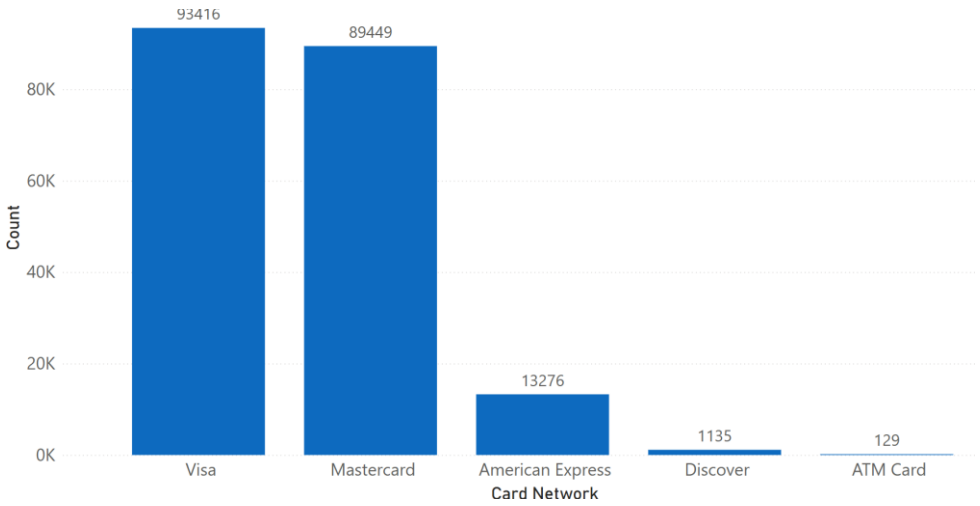

Fig. 17 Top 5 Compromised Card Networks by Count in 2023

6 Discussion

Our study examines how stolen personal information is organized, shared, and distributed within Telegram's messaging environment, addressing a critical gap in research on identity theft dynamics in such messaging platforms. The findings provide important insights into how these platforms have transformed criminal ecosystems and offer new perspectives on the theoretical frameworks used to understand cybercrime.

6.1 Key Findings

Our analysis directly addresses the three research questions posed at the outset of this study, revealing significant insights into how Telegram facilitates identity theft operations and the patterns of criminal activity within these ecosystems. The findings demonstrate that Telegram has evolved into a "Transformative Criminal Convergence Space" that fundamentally redefines how identity theft operations function compared to traditional cybercrime environments.

The platform's architecture systematically enables criminal activity through three key mechanisms. First, it dramatically expands motivated offenders through its user-friendly interface that requires minimal technical expertise. Second, it centralizes suitable targets through broadcast functionality that instantly distributes stolen credentials to thousands of subscribers. Third, it weakens capable guardianship through minimal registration requirements and technical features that impede traditional law enforcement monitoring. While these findings align with routine activity theory's

convergence of offenders, targets, and weak guardianship (refer to Section 3.2), we found that Telegram differs significantly from traditional darknet markets in terms of criminal supply chain and operational dynamics.

This architectural advantage positions Telegram as a key intermediary in the underground supply chain, facilitating the rapid movement of compromised data from initial breach sources to end-user criminals engaged in fraud operations. The platform effectively democratizes access to stolen credentials, expanding the pool of potential fraud actors beyond technically sophisticated criminals to include opportunistic offenders who lack the skills to navigate traditional darknet infrastructures.

Having established how Telegram's architecture enables criminal convergence, our analysis of data sharing patterns reveals clear criminal preferences and strategic priorities. Financial credentials vastly outnumber other forms of PII, with credit card information comprising 82% of all compromised data. This dominance supports the low-risk/high-return criminological models that characterize modern cybercrime and reflects clear criminal priorities focused on immediately monetizable information. The relative scarcity of Social Security Numbers (0.01%) compared to credit card details suggests that Telegram channels serve a specific niche within the broader identity theft ecosystem, focusing on rapid-turnover financial fraud rather than comprehensive identity reconstruction. This specialization pattern demonstrates how the platform's accessibility enables basic financial fraud while more sophisticated identity theft operations requiring government-issued identifiers operate through different, more exclusive channels.

Beyond data type preferences, our analysis reveals sophisticated temporal coordination among criminal operators. The distinct peaks in June, January, and March, demonstrate how channel operators coordinate data releases to maximize market impact during periods of heightened consumer activity. This synchronization across different PII types indicates that Telegram channels serve as distribution networks for major data breaches rather than continuous collection points, revealing how the platform's broadcast capabilities enable large-scale, coordinated criminal activity that would be difficult to achieve through other means.

The strategic nature of these operations becomes further evident when examining victimization patterns across different banks, card types, and geographic regions, revealing that non-random targeting that goes beyond simple population correlation. The concentration in Texas, California, and Florida, combined with the disproportionate targeting of major financial institutions like Bank A and Bank B, indicates sophisticated criminal intelligence gathering and strategic victim selection. The alignment between payment network compromise rates and their market prevalence suggests that criminals are effectively exploiting the most used financial instruments rather than seeking specific vulnerabilities. However, the quality issues identified, with only 25% of addresses being geocodable, reveal how the platform's accessibility comes at the cost of data reliability, creating inefficiencies that may limit criminal effectiveness while providing detection opportunities for security professionals.

6.2 Theoretical Implications

Our findings have significant implications for criminological theory, particularly Routine Activity Theory (Cohen and Felson, 1979) [1]. The patterns we documented support our conceptualization of Telegram as a "criminal convergence space" where the three elements of crime, motivated offenders, suitable targets, and absence of capable guardianship, interact in unprecedented ways. Unlike darknet markets that require technical expertise to access, creating a form of guardianship through complexity, Telegram's combination of accessibility and anonymity creates an environment that lowers barriers for motivated offenders while simultaneously raising barriers against law enforcement intervention.

The platform's architecture enables what we term a "broadcast marketplace" model, where sellers can efficiently distribute stolen data to large audiences while maintaining operational security. This represents a fundamental shift from traditional darknet markets that require technical expertise to access. Telegram's surface web accessibility, combined with its hybrid encryption features, creates optimal conditions for criminal convergence by reducing guardianship through both technical barriers to law enforcement and the platform's semi-encrypted nature, while simultaneously providing easy access for motivated offenders to connect with suitable targets.

The temporal synchronization observed across different PII types extends theoretical understanding of how digital platforms transform criminal opportunities by demonstrating that accessibility and speed can be more important than maximum security for many criminal purposes. The coordination of complete identity packages suggests that criminal networks use these channels to rapidly spread recently acquired data, taking advantage of the platform's unique combination of reach and anonymity to create new forms of criminal convergence that were previously impossible.

Our research also contributes to platform affordance theory (Bucher and Helmond, 2018) [53] to criminal contexts by demonstrating how Telegram's specific features like public channels, large group capacity, and surface web accessibility, shape criminal behavior in ways distinct from both traditional darknet markets and other messaging platforms like Signal or WhatsApp. The quality issues identified, with only 25% of addresses being geocodable, reveal how this accessibility comes at the cost of data reliability, creating inefficiencies that may limit criminal effectiveness while providing detection opportunities for security professionals.

6.3 Practical Implications

The implications of these findings extend across multiple stakeholder groups. Understanding how Telegram's architecture facilitates criminal convergence provides law enforcement and cybersecurity professionals with crucial insights for developing monitoring and intervention strategies. The rapid pace of content publication and deletion, combined with the volume of low-quality information, suggests that automated monitoring systems must be capable of real-time processing and intelligent filtering

to extract actionable intelligence. The identification of temporal patterns in criminal activity provides predictive intelligence that can inform dynamic response strategies.

Financial institutions should recognize that the identified seasonal patterns provide predictive intelligence for fraud prevention systems. The concentration of criminal attention on major banks suggests that size and visibility create risk, requiring these institutions to implement sophisticated detection and prevention measures. The temporal patterns could inform dynamic risk assessment models that adjust fraud detection sensitivity based on historical criminal activity cycles.

The geographic patterns can inform targeted awareness campaigns and resource allocation strategies. The disproportionate victimization in Texas compared to California, despite population differences, suggests that regional factors beyond demographics influence targeting and warrant investigation. This geographic intelligence can help direct both prevention efforts and victim services to areas of highest risk.

This research provides the foundation for three practical applications that could significantly enhance identity theft prevention and response. Real-time monitoring systems could be developed to alert financial institutions when specific PII formats or institutional targeting patterns are detected, enabling preventive measures before widespread fraud occurs. Victim notification platforms could be designed to feed web-based services that proactively inform individuals when their credentials appear in criminal channels (Caitlyn et al. 2025) [54]. Enhanced risk modelling could help insurers and credit agencies refine their assessment frameworks around digital financial fraud based on the identified patterns.

6.4 Methodological Limitations

Our methodology consists of several important limitations which reflect the inherent challenges in studying criminal ecosystems within messaging environments. However, the systematic approach, validation mechanisms, and analytical framework provide a foundation for understanding identity theft operations on Telegram while acknowledging these constraints.

Platform Specificity : The focus on Telegram limits the generalizability of findings to other messaging platforms due to fundamental architectural differences. Platforms like WhatsApp and Signal differ in several key aspects: they lack Telegram's public channel functionality, have more restrictive group size limits, and implement different encryption protocols. While we expect some operational similarities in criminal communication patterns across platforms, the specific patterns observed on Telegram may not directly translate to these alternative platforms due to their architectural differences.

Sampling Limitations: Despite systematic channel identification procedures, our sample represents an unknown fraction of the total criminal ecosystem on Telegram. Channels with more restrictive access controls or those operating in languages other than English were underrepresented in our sample, potentially biasing findings toward more accessible criminal operations.

Data Quality Issues: As noted in our validation procedures, much of the data circulating in criminal channels exhibits quality issues, including deliberate falsification, partial information, and formatting inconsistencies. While our validation mechanisms identify many of these issues, they inevitably affect the completeness and reliability of certain analyses.

Ethical Boundaries: Our commitment to ethical research practices limited certain forms of data collection and analysis. We deliberately avoided active participation in criminal channels, restricted analysis to publicly accessible information, and implemented data security protocols that may have constrained analytical possibilities.

6.5 Ethical Considerations

These ethical considerations were integral to our methodological decisions, reflecting the complex balance required when studying criminal activities in digital environments. While the collected data originated from public channels where it was already exposed, we implemented strict security protocols to prevent secondary victimization. Everyone involved in the collection, scraping, or viewing of the data had signed a non-disclosure agreement (NDA) for the protection and privacy of the compromised information.

6.6 Policy and Regulatory Considerations

The research demonstrates that while anonymity provides legitimate privacy benefits, the accessibility of platforms like Telegram creates new challenges for preventing criminal exploitation. However, our findings also suggest that heavy-handed regulatory approaches might be counterproductive, as criminals could simply migrate to more technically sophisticated platforms that would be harder to monitor.

The quality issues identified in criminal communications suggest that current approaches to platform regulation may be missing important opportunities. Rather than focusing solely on content removal, regulatory frameworks could consider approaches that increase data friction and reduce criminal efficiency without compromising legitimate privacy needs.

7 Limitations

While this research presents a comprehensive methodology for analyzing stolen personal data circulated on Telegram, several important limitations must be acknowledged.

1. *Temporal and Scale Constraints*: The study examines only 25 Telegram channels during a limited period (2023), which may not provide longitudinal insights into

evolving cybercriminal behavior patterns or capture seasonal variations in criminal activities beyond the observed timeframe.

2. *Data comprehensiveness*: The research lacks cross-validation with known data breaches and by concentrating primarily on text-based PII while excluding image-based data, the study may underestimate the full scope of compromised information being shared. Our approach does not effectively address sophisticated evasion techniques that criminals might employ, such as image-based text, coded language, or encryption of sensitive fields within messages.

3. *Technical Dependencies*: Our methodology relies on GUI-based scraping through Sikuli IDE, which is inherently fragile to interface changes, while also depending heavily on third-party services (AWS Comprehend, Google Maps Geocoding API, etc.) that impose rate limits. These dependencies can cause points of failure in the data processing pipeline and may constrain the scale and frequency of data collection.

4. *Privacy concerns*: The collection and processing of actual victims' PII, even for research purposes, raises significant privacy concerns. It can create tension between public interest and individual rights, particularly when considering future victim notification systems that could potentially re-traumatize individuals.

   Note: In accordance with ethical research and privacy protection protocols, the scraped JSON files containing sensitive information have been deliberately withheld from sharing to uphold our commitment to preventing secondary victimization.

5. *Analytical Precision*: NLP techniques inevitably produce both false positives and false negatives when processing informal, jargon-filled criminal communications. For instance, only 25% of collected addresses were proven valid after geocoding, highlighting the challenges of analyzing deliberately obfuscated criminal data.

Despite these limitations, this research establishes a foundational framework for understanding criminal operations on instant messaging platforms and provides valuable insights into the evolving landscape of digital identity theft. While our findings may not capture the complete scope of criminal activity on Telegram, they offer significant empirical evidence about data distribution patterns, geographic targeting strategies, and operational characteristics that can inform both theoretical advancement and practical countermeasures in cybersecurity and digital criminology.

8 Conclusion and Future Directions

This study demonstrates the alarming scale at which cybercriminals are exploiting the surface-accessible messaging platforms, particularly Telegram, to traffic in stolen personal and financial information. Our study highlights the operation of an illicit supply chain on these surface platforms. We present a multi-stage automated methodology combining GUI automation, NLP and specialized APIs confirming that

it can effectively collect and structure PII at scale from these platforms. Our analysis revealed that credit card information dominates the compromised data, confirming the hypothesis about the types of PII most valued by cybercriminals and supporting low-risk/high-return criminological models. The findings demonstrate that larger financial institutions experience significantly higher volumes of compromised cards, with clear geographic patterns correlating to population centers. Bank A and Bank B emerged as the most targeted institutions, while Texas, California, and Florida showed the highest concentration of victims.

This research validates the potential of automated methodologies to mitigate identity theft by establishing patterns and vulnerabilities that would be impossible to detect manually. Our approach provides security professionals with actionable intelligence while laying groundwork for more sophisticated measures.

Several promising avenues for future research emerge from this work. Future studies could explore the development of real-time notification systems built upon similar data collection frameworks to alert individuals when their personal information appears in known criminal channels, necessitating secure verification protocols, false alarm prevention filters and appropriate support resources. Additionally, methods to extract image-based PII, such as that found on driver's licenses and checks, from identified Telegram channels could be explored, along with adapting these methodologies to other messaging platforms like Discord, Signal and WhatsApp through specialized scrapers. Furthermore, leveraging the temporal and geographic patterns identified in our research, predictive analytics could be developed to forecast the surges in criminal activity and identify the emerging targets. Finally, a secure collaborative platform could be established where financial institutions, law enforcement agencies and cybersecurity researchers could facilitate sharing of insights, countermeasures and threat intelligence.

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Appendix A API Services and Privacy Specifications

Service	Description	Privacy & Security
AWS Comprehend	Amazon Web Services Comprehend, a natural language processing (NLP) based service was utilized to identify and categorize PII within the collected dataset. The service provided automated detection of sensitive data types including names, addresses, phone numbers, and financial account information, enabling systematic classification of compromised credentials across the dataset [50].	Implements end-to-end encryption and automatically deletes input data after processing. The service operates in isolated environments with strict access controls and maintains SOC 2 Type II and GDPR compliance. All processing occurs within designated AWS regions with comprehensive audit logging for accountability..
BINChecker API	Bank Identification Number (BIN) Checker API services were employed to validate and categorize credit card information found in the dataset. This API enabled the identification of issuing banks, card types, and geographic regions associated with compromised payment cards. The BIN checking process provided intelligence for institutional analysis and helped verify the authenticity of compromised financial data [51].	Uses HTTPS encryption and stateless architecture with no persistent data storage. Query logs are purged within 24 hours, and the service maintains PCI DSS Level 1 compliance. Access is controlled through API key authentication with rate limiting and detailed audit trails.
Google Maps Geocoding API	While APIs from providers like OpenStreetMap, Mapbox, and Here also deliver robust features, Google's reputation for unmatched accuracy, extensive data coverage, and a broad range of a$_4$d$_6$ditional services frequently makes it the preferred choice for this research as it demands precise location data for geographic analysis of victim distribution patterns [52].	Employs TLS 1.2+ encryption and processes queries without retaining PII beyond immediate operations. The service maintains ISO 27001 and SOC 2 compliance, supports data residency controls, and uses anonymized processing pipelines that separate location data from personal identifiers.

Appendix B Data Visualization Tools and Platforms

All graphs, charts, and maps presented in this study were generated using industry-standard business intelligence and data visualization platforms to ensure accuracy, clarity and professional presentation of research findings.

Power BI : Microsoft Power BI was utilized for creating interactive dashboards and statistical visualizations including bar charts, line graphs, and pie charts displaying temporal patterns, categorical distributions, and comparative analyses across different data dimensions.

Tableau: Tableau was employed for advanced data visualization and geographic mapping, particularly for creating the state-level distribution maps and complex multi-dimensional visualizations.

Appendix C Financial Institution Anonymization and Complete List

Anonymity of Banks and Other Financial Institutions: To protect institutional reputations and avoid singling out specific organizations, all banks and financial institutions referenced in this study have been anonymized using generic identifiers (Bank A, Bank B, Organization A, etc.). This approach ensures that the research findings focus on systemic patterns rather than individual institutional vulnerabilities, while maintaining the analytical integrity of the data.

The following financial institutions were identified in the dataset and are listed in alphabetical order (not reflective of the order or frequency presented in the paper's analysis):

- Ally Bank
- Ameris Bank
- American Express National Bank
- Arvest Bank
- Associated Bank National Association
- Atlantic Union Bank
- Bank of America National Association
- Bank of Hawaii
- Bank of Montreal Bank NA
- Bank of New York Mellon
- Bank of New York Mellon National Association
- Bank of Oklahoma Kansas and Texas National Association
- Bank of the West
- Banc of California
- BankUnited National Association
- Barclays Bank Delaware

- Cadence Bank
- Canadian Imperial Bank of Commerce Bank United States of America
- Capital One National Association
- Cathay Bank
- Central Trust Bank
- Centennial Bank
- Chase Bank (JPMorgan Chase Bank National Association)
- Citibank National Association
- Citizens Bank National Association
- City National Bank
- City National Bank of Florida
- Comerica Bank
- Commerce Bank
- Customers Bank
- Deutsche Bank Trust Company Americas
- Discover Bank
- East West Bank
- Eastern Bank
- EverBank National Association
- Fifth Third Bank National Association
- First Data Corporation
- First Citizens Bank & Trust Company
- First Financial Bank
- First Hawaiian Bank
- First Horizon Bank
- First Interest Bank
- First Merchants Bank
- First National Bank of Omaha
- First National Bank of Pennsylvania
- First-Citizens Bank & Trust Company
- FirstBank
- Flagstar Bank National Association
- Frost Bank
- Fulton Bank National Association
- Glacier Bank
- Goldman Sachs Bank United States of America
- Hancock Whitney Bank
- Heartland Financial Trust Fund Bank
- Hong Kong and Shanghai Banking Corporation Bank United States of America National Association
- Huntington National Bank
- ING Bank
- Independent Financial
- JPMorgan Chase Bank National Association
- KeyBank National Association

- Manufacturers & Traders Trust Company
- Merchants Bank of Indiana
- Morgan Stanley Bank National Association
- Morgan Stanley Private Bank National Association
- Northern Trust Company
- Old National Bank
- Pacific Premier Bank National Association
- Pennsylvania National Corporation Bank National Association
- Pinnacle Bank
- Raymond James Bank
- Regions Bank
- Renasant Bank
- Rockland Trust
- Santander Bank National Association
- ServisFirst Bank
- Simmons Bank
- Social Finance Bank National Association
- SouthState Bank National Association
- State Street Bank & Trust Company
- Stifel Bank & Trust
- Synovus Bank
- Texas Capital Bank
- Toronto-Dominion Bank National Association
- TowneBank
- TriState Capital Bank
- Truist Bank
- Trustmark National Bank
- Umpqua Bank
- Union Bank & Trust Company
- United Bank
- United Community Bank
- United Missouri Bank National Association
- US Bank
- Valley National Bank
- Washington Federal Bank
- Webster Bank National Association
- Wells Fargo Bank National Association
- WesBanco Bank
- Western Alliance Bank
- Zions Bancorporation National Association

Appendix D Source Code and Technical Resources

The source code for all text parsers and analytical tools used in this study is publicly available in the GitHub repository linked on the title page to maintain reviewer anonymity.

Appendix E Detailed Flowchart for Data Scraping

```
Start
├─ Open Sikuli IDE
├─ Import libraries and modules
├─ Create new script
├─ Open Google Chrome browser
├─ Look for Telegram Web
│  ├─ Open a new tab
│  ├─ Navigate to "web.telegram.org"
├─ Read channel names from a text file
│  ├─ Open the text file
│  ├─ Read line by line
│  └─ Store each channel name
├─ Iterate over each channel
│  ├─ Search for the channel
│  │  ├─ Locate search field
│  │  ├─ Enter channel name
│  │  └─ Click search button
│  ├─ Open the channel
│  │  ├─ Click on the channel from search results
│  │  ├─ Join the channel
│  │  └─ Click on channel info
│  │     ├─ Click on vertical three dots
│  │     ├─ Click on "Export Channel History"
│  │     ├─ Choose content to export
│  │     │  ├─ Select photos
│  │     │  ├─ Select textual messages
│  │     │  └─ Confirm content selection
│  │     ├─ Set size limit of chosen content
│  │     │  ├─ Specify maximum file size
│  │     │  └─ Confirm size limit
│  │     ├─ Choose export format
│  │     │  ├─ Select format (e.g., JSON or HTML)
│  │     │  └─ Confirm export format
│  │     ├─ Choose download path
│  │     │  ├─ Select local system directory
│  │     │  └─ Confirm download path
│  │     ├─ Choose date range
│  │     │  ├─ Select from date
│  │     │  ├─ Select to date
│  │     │  └─ Confirm date range
│  │     └─ Click on "Export" button
├─ Click on "Show My Data."
└─ End
```